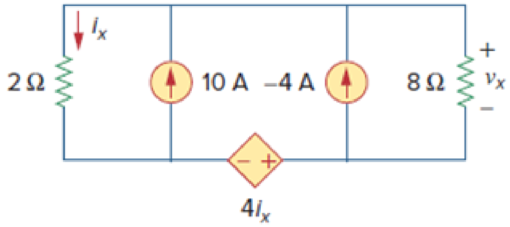
Chapter 4, Problem 19P

Problem

Use superposition to solve for *vx* in the circuit of Fig. 4.87.

Figure. 4.87:



Step-by-step solution

Step 1 of 11

Refer to circuit diagram in Figure 4.87 in the text book.

The circuit has two independent sources, assume that the output voltage is,

$$v_x = V_1 + V_2$$

Here, V_1 and V_2 are the voltage responses due to -4 A current source and 10 A current source respectively.

To obtain V_1, set the 10 A current source to zero, the modified circuit is shown in Figure 1.

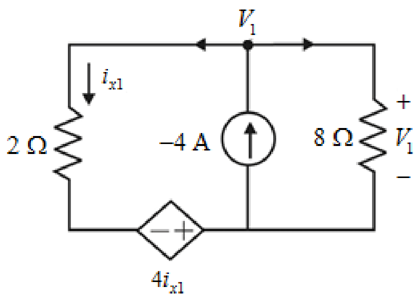


Figure 1

Step 2 of 11

From the circuit shown in Figure 1, the current i_{x1} is,

$$i_{x1} = \frac{V_1 - (-4i_{x1})}{2}$$
$$2i_{x1} = V_1 + 4i_{x1}$$
$$-2i_{x1} = V_1$$
$$i_{x1} = -\frac{V_1}{2} \quad \text{...... (1)}$$

Apply nodal analysis at node V_1.

$$\frac{V_1 - (-4i_{x1})}{2} + \frac{V_1}{8} = -4$$

Substitute $-\frac{V_1}{2}$ for i_{x1}.

$$\frac{V_1 + 4\left(-\frac{V_1}{2}\right)}{2} + \frac{V_1}{8} = -4$$
$$\frac{V_1 - 2V_1}{2} + \frac{V_1}{8} = -4$$
$$\frac{-4V_1 + V_1}{8} = -4$$
$$-3V_1 = -32$$
$$V_1 = \frac{32}{3} \text{ V}$$

Step 3 of 11

To obtain V_2, set the -4 A current source to zero, the modified circuit is shown in Figure 2.

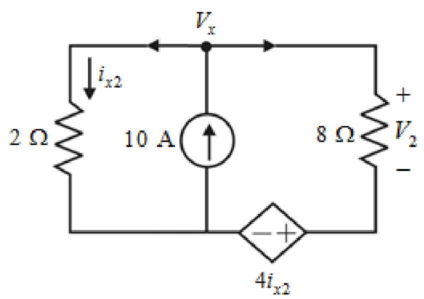


Figure 2

Step 4 of 11

From the circuit shown in Figure 2, the current i_{x2} is,

$$i_{x2} = \frac{V_x}{2} \quad \text{...... (2)}$$

Apply nodal analysis at node V_x.

$$\frac{V_x}{2} + \frac{V_x - 4i_{x2}}{8} = 10$$

Substitute $\frac{V_x}{2}$ for i_{x2}.

$$\frac{V_x}{2} + \frac{V_x - 4\left(\frac{V_x}{2}\right)}{8} = 10$$

$$\frac{V_x}{2} + \frac{V_x - 2V_x}{8} = 10$$

$$\frac{4V_x - V_x}{8} = 10$$

$$3V_x = 80$$

$$V_x = \frac{80}{3}$$

Step 5 of 11

Apply Kirchhoff's voltage law to the right side loop.

$$-V_x + V_2 + 4i_{x2} = 0$$

Substitute $\frac{V_x}{2}$ for i_{x2}.

$$-V_x + V_2 + 4\left(\frac{V_x}{2}\right) = 0$$

$$-V_x + V_2 + 2V_x = 0$$

$$V_2 + V_x = 0$$

$$V_2 = -V_x$$

$$V_2 = -\frac{80}{3}\ \text{V}$$

Now, calculate the voltage v_x due to both sources acting.

$$v_x = V_1 + V_2$$

Substitute $\frac{32}{3}$ V for V_1, and $-\frac{80}{3}$ V for V_2.

$$\begin{aligned} v_x &= \frac{32}{3} - \frac{80}{3}\ \text{V} \\ &= -\frac{48}{3}\ \text{V} \\ &= -16\ \text{V} \end{aligned}$$

Therefore, the voltage v_x in the circuit is $\boxed{-16\ \text{V}}$.

Step 6 of 11

ORCAD simulation:

Let the $-4\ \mathrm{A}$ current source be active and disconnect the $10\ \mathrm{A}$ current source.

Construct the circuit diagram in ORCAD as shown in Figure 3.

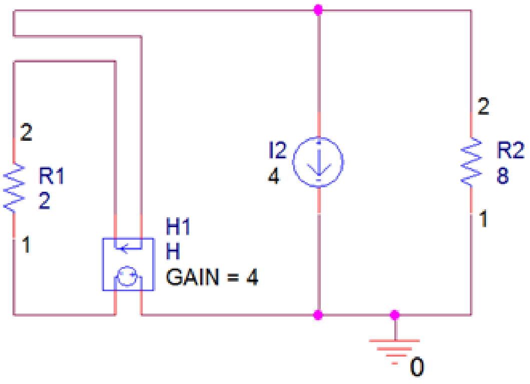


Figure 3

Step 7 of 11

Go to Pspice and create a new simulation profile. Edit the simulations profile and select 'Bias point' simulation as shown in Figure 4. Click on ok.

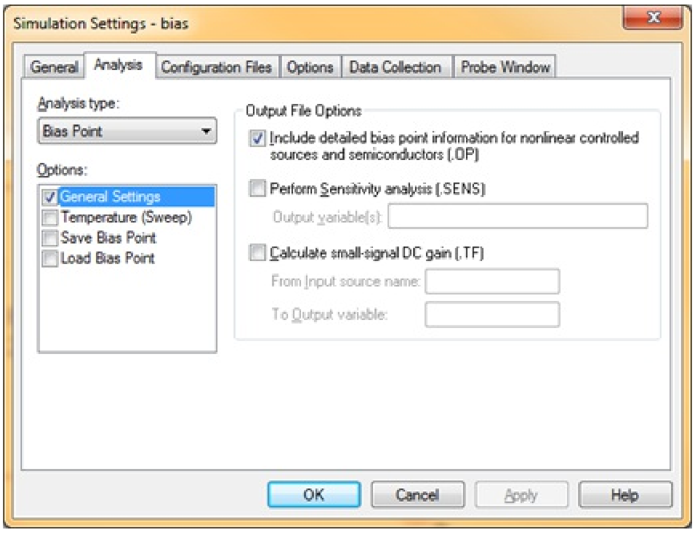


Figure 4

Step 8 of 11

Click on voltage bias simulation display. Press F11 to simulate the circuit.

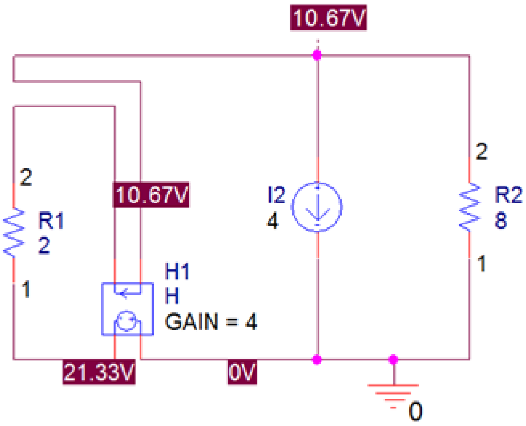


Figure 5

The value of the voltage v_{x1} due to -4 A current source is 10.67 V.

Step 9 of 11

Let the 10 A current source be active and disconnect the -4 A current source.

Construct the circuit diagram in ORCAD as shown in Figure 6.

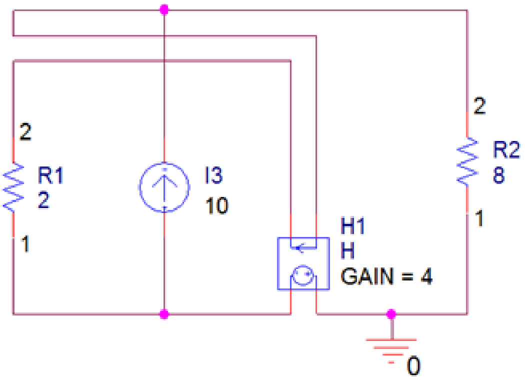


Figure 6

Step 10 of 11

Click on voltage bias simulation display. Press F11 to simulate the circuit.

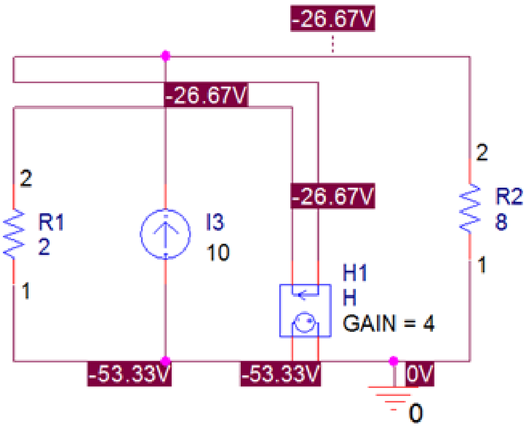


Figure 7

The value of the voltage v_{x2} due to 10 A current source is -26.67 V.

Step 11 of 11

Calculate the value of the voltage v_x due to both sources activated.

$$v_x = v_{x1} + v_{x2}$$

Substitute 10.67 V for v_{x1} and -26.66 V for v_{x2}.

$$\begin{aligned} v_x &= 10.67 - 26.67\text{ V} \\ &= -16\text{ V} \end{aligned}$$

Therefore, the value of the voltage, v_x due to both sources activated is $\boxed{-16\text{ V}}$.